Sales Data Analysis with Python

A part of Big Data Analysis

Restaurant Sales Dataset

Importing the required Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Importing the dataset

```
df = pd.read_csv('Sales.csv')
df.head(10)
```

```
   Unnamed: 0  Unnamed: 1  Unnamed: 2  Unnamed: 3  Unnamed: 4  Unnamed: 5  Unnamed: 6
0         NaN         NaN         NaN         NaN         NaN         NaN         NaN
1         NaN         NaN         NaN         NaN         NaN         NaN         NaN
2         NaN         NaN         NaN         NaN         NaN         NaN         NaN
3         NaN         NaN         NaN         NaN         NaN         NaN         NaN
4         NaN         NaN         NaN         NaN         NaN         NaN         NaN
5         NaN         NaN         NaN         NaN         NaN         NaN         NaN
6         NaN         NaN         NaN         NaN         NaN         NaN         NaN
7         NaN         NaN         NaN         NaN         NaN         NaN         NaN
8         NaN         NaN         NaN         NaN         NaN         NaN         NaN
9         NaN         NaN         NaN         NaN         NaN         NaN         NaN
   Unnamed: 7  Unnamed: 8  Unnamed: 9  Unnamed: 10  Unnamed: 11  Unnamed: 12  Unnamed: 13
0         NaN         NaN         NaN          NaN          NaN          NaN          NaN
1         NaN         NaN         NaN          NaN          NaN          NaN          NaN
2         NaN         NaN         NaN          NaN          NaN          NaN          NaN
3         NaN         NaN         NaN          NaN          NaN          NaN          NaN
4         NaN         NaN         NaN          NaN          NaN          NaN          NaN
5         NaN         NaN         NaN          NaN          NaN          NaN          NaN
6         NaN         NaN         NaN          NaN          NaN          NaN          NaN
7         NaN         NaN         NaN          NaN          NaN          NaN          NaN
8         NaN         NaN         NaN          NaN          NaN          NaN          NaN
9         NaN         NaN         NaN          NaN          NaN          NaN          NaN
   Unnamed: 14
0          NaN
1          NaN
2          NaN
3          NaN
4          NaN
5          NaN
6          NaN
7          NaN
8          NaN
9          NaN
```

Drop Unwanted Column

```
df.drop(['Unnamed: 0', 'Unnamed: 1', 'Unnamed: 2', 'Unnamed: 3', 'Unnamed: 4', 'Unnamed: 5', 'Unnamed: 6', 'Unnamed: 7', 'Unnamed: 8', 'Unnamed: 9', 'Unnamed: 10', 'Unnamed: 11', 'Unnamed: 12', 'Unnamed: 13', 'Unnamed: 14'], axis=1, inplace=True)
```

Checking the dataset details

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 10):
 #   Order ID        9800 non-null   object
 #   Product         9800 non-null   object
 #   Quantity Ordered 9800 non-null  object
 #   Price Each      9800 non-null   object
 #   Order Date      9800 non-null   object
 #   Purchase Address 9800 non-null  object
 #   Month           9800 non-null   object
 #   Sales           9800 non-null   object
 #   City            9800 non-null   object
 #   Hour            9800 non-null   object
dtypes: object(10)
memory usage: 765.7+ KB
```

```
df.shape
```

```
(9800, 10)
```

```
df.describe()
```

```
       Order ID  Product  Quantity Ordered  Price Each  Order Date  Purchase Address  Month  Sales   City  Hour
count      9800     9800              9800        9800        9800              9800   9800   9800   9800  9800
unique     9800       19                 9          79        5337              9131     12   5337     10    24
top        NaN      NaN               NaN         NaN         NaN               NaN    NaN    NaN    NaN   NaN
freq        NaN      NaN               NaN         NaN         NaN               NaN    NaN    NaN    NaN   NaN
```

Change the column name

```
df.rename(columns={'Order ID':'Order_ID', 'Product':'Product', 'Quantity Ordered':'Quantity_Ordered', 'Price Each':'Price_Each', 'Order Date':'Order_Date', 'Purchase Address':'Purchase_Address', 'Month':'Month', 'Sales':'Sales', 'City':'City', 'Hour':'Hour'})
```

```
Index(['Order_ID', 'Product', 'Quantity_Ordered', 'Price_Each', 'Order_Date',
       'Purchase_Address', 'Month', 'Sales', 'City', 'Hour'],
      dtype=object)
```

Change the datatype series

```
df['Order_ID'] = df['Order_ID'].astype('int')
df['Quantity_Ordered'] = df['Quantity_Ordered'].astype('int')
df['Price_Each'] = df['Price_Each'].astype('float')
df['Month'] = df['Month'].astype('int')
df['Sales'] = df['Sales'].astype('float')
df['Hour'] = df['Hour'].astype('int')
df.dtypes
```

```
Order_ID            int64
Product            object
Quantity_Ordered    int64
Price_Each        float64
Order_Date         object
Purchase_Address   object
Month              int64
Sales             float64
City               object
Hour               int64
dtype: object
```

Check for missing values

```
df.isnull()
```

```
      Order_ID  Product  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
0        False    False             False       False       False             False  False  False  False  False
1        False    False             False       False       False             False  False  False  False  False
2        False    False             False       False       False             False  False  False  False  False
3        False    False             False       False       False             False  False  False  False  False
4        False    False             False       False       False             False  False  False  False  False
5        False    False             False       False       False             False  False  False  False  False
6        False    False             False       False       False             False  False  False  False  False
7        False    False             False       False       False             False  False  False  False  False
8        False    False             False       False       False             False  False  False  False  False
9        False    False             False       False       False             False  False  False  False  False
...        ...      ...               ...         ...         ...               ...    ...    ...    ...    ...
9790     False    False             False       False       False             False  False  False  False  False
9791     False    False             False       False       False             False  False  False  False  False
9792     False    False             False       False       False             False  False  False  False  False
9793     False    False             False       False       False             False  False  False  False  False
9794     False    False             False       False       False             False  False  False  False  False
9795     False    False             False       False       False             False  False  False  False  False
9796     False    False             False       False       False             False  False  False  False  False
9797     False    False             False       False       False             False  False  False  False  False
9798     False    False             False       False       False             False  False  False  False  False
9799     False    False             False       False       False             False  False  False  False  False

[9800 rows x 10 columns]
```

```
df.isnull().sum()
```

```
Order_ID            0
Product             0
Quantity_Ordered    0
Price_Each          0
Order_Date          0
Purchase_Address    0
Month               0
Sales               0
City                0
Hour                0
dtype: object
```

Remove the duplicate records

```
df.drop_duplicates(keep='first')
```

```
      Order_ID  Product  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
0       141234   iPhone                 1         700  01/22/2019  942 Johnston St,...      1  700.  Boston    8.
1       141235  Lightn                 1         14.  01/28/2019  61 Hill St, Dall...      1  14.  Dallas   14.
2       141236  Wired                  1         11.  01/17/2019  333 8th St, Los ...      1  11.  Los ...  14.
3       141237  27in"                  1         150  01/05/2019  20 Hill St, Los ...      1  150  Los ...  11.
4       141238  Wired                  1         11.  01/18/2019  778 Maple St, Bo...      1  11.  Boston   19.
5       141239  AAA                    1         2.9  01/29/2019  214 Chestnut St,...      1  2.9  Atlanta   11.
6       141240  Screen                 1         109  01/26/2019  070 Park St, Los ...      1  109  Los ...   7.
7       141241  Wired                  1         11.  01/10/2019  779 King St, Bos...      1  11.  Boston   12.
8       141242  iPhone                 1         700  01/26/2019  657 Hill St, Dal...      1  700  Dallas    10.
9       141243  Lightn                 1         14.  01/12/2019  636 3th St, New ...      1  14.  New ...  12.
...        ...     ...               ...         ...         ...               ...    ...    ...    ...    ...
9790    259353  AAA                    1         2.9  09/17/2019  840 Highland St,...      9  2.9  Boston   16.
9791    259354  iPhone                 1         700  09/01/2019  216 Dogwood St, ...      9  700  Dallas    7.
9792    259355  iPhone                 1         700  09/23/2019  220 12th St, San...      9  700  San ...   1.
9793    259356  34in"                  1         325  09/19/2019  11 Park St, Los ...      9  325  Los ...   22.
9794    259357  Flatscr                1         300  09/30/2019  837 Chestnut St,...      9  300  Boston    9.
9795    259358  LGB30                  1         71.  09/29/2019  135 10th St, Atl...      9  71.  Atlanta   14.
9796    259359  Wired                  1         11.  09/07/2019  708 Back St, Dal...      9  11.  Dallas    19.
9797    259360  LGB30                  1         71.  09/12/2019  75 Learning St, ...      9  71.  Los ...   15.
9798    259361  34in"                  1         325  09/12/2019  854 Walnut St, B...      9  325  Boston    20.
9799    259362  AAA                    1         2.9  09/30/2019  261 10th St, San...      9  2.9  San ...   16.

[9800 rows x 10 columns]
```

Consider the attribute of columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 10):
 #   Order_ID        9800 non-null   int64
 #   Product         9800 non-null   object
 #   Quantity_Ordered 9800 non-null  int64
 #   Price_Each      9800 non-null   float64
 #   Order_Date      9800 non-null   object
 #   Purchase_Address 9800 non-null  object
 #   Month           9800 non-null   int64
 #   Sales           9800 non-null   float64
 #   City            9800 non-null   object
 #   Hour            9800 non-null   int64
dtypes: int64(4), float64(3), object(3)
memory usage: 765.7+ KB
```

```
df.shape
```

```
(9800, 10)
```

```
df['Product'].value_counts()
```

```
Product
USB-C Charging Cable    1000
Lightning Charging C    1000
AAA Batteries (4-pack)  1000
AA Batteries (4-pack)   1000
Wired Headphones        1000
Apple Airpods Headpho    800
Apple Airpods Headpho    800
Flatscr                  500
27in 4K Gaming Monitor   500
Bose SoundSport Head     500
34in Ultra Wide Monitor  500
Google Phone             500
Vareebadd phone          500
LG Washing Machine       500
LG Dryer                 500
Lightning Charging C     500
Bose SoundSport Head     500
iPhone                   500
Screen                   500
Wired                    500
Name: count, dtype: int64
```

```
df['Quantity_Ordered'].value_counts()
```

```
Quantity_Ordered
1    8800
2     500
3     100
4     100
5     100
6     100
7     100
8     100
9     100
Name: count, dtype: int64
```

```
df['Price_Each'].value_counts()
```

```
Price_Each
11.95    1000
14.95    1000
2.99     1000
150      500
109      500
700      500
11.95    500
71.95    500
300      500
325      500
14.95    500
2.99     500
11.95    500
700      500
109      500
71.95    500
325      500
300      500
11.95    500
2.99     500
Name: count, dtype: int64
```

```
df['Order_Date'].value_counts()
```

```
Order_Date
12/31/2019 23:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
12/31/2019 19:59    1000
Name: count, dtype: int64
```

```
df['Purchase_Address'].value_counts()
```

```
Purchase_Address
Google Phone                    500
Vareebadd phone                 500
LG Washing Machine              500
LG Dryer                        500
Lightning Charging C            500
Bose SoundSport Head            500
iPhone                          500
Screen                          500
Wired                           500
USB-C Charging Cable            500
Lightning Charging C            500
AAA Batteries (4-pack)          500
AA Batteries (4-pack)           500
Wired Headphones                500
Apple Airpods Headpho           500
Apple Airpods Headpho           500
Flatscr                         500
27in 4K Gaming Monitor          500
Bose SoundSport Head            500
34in Ultra Wide Monitor         500
Google Phone                    500
Vareebadd phone                 500
LG Washing Machine              500
LG Dryer                        500
Lightning Charging C            500
Bose SoundSport Head            500
iPhone                          500
Screen                          500
Wired                           500
Name: count, dtype: int64
```

```
df['Month'].value_counts()
```

```
Month
12    1000
11    1000
10    1000
9     1000
8     1000
7     1000
6     1000
5     1000
4     1000
3     1000
2     1000
1     1000
Name: count, dtype: int64
```

```
df['Sales'].value_counts()
```

```
Sales
11.95    1000
14.95    1000
2.99     1000
150      500
109      500
700      500
11.95    500
71.95    500
300      500
325      500
14.95    500
2.99     500
11.95    500
700      500
109      500
71.95    500
325      500
300      500
11.95    500
2.99     500
Name: count, dtype: int64
```

```
df['City'].value_counts()
```

```
City
San Francisco  2000
Los Angeles    1000
Boston         1000
Dallas         1000
Atlanta        1000
New York City  1000
Seattle        1000
Portland       1000
Austin         1000
Portland       1000
Name: count, dtype: int64
```

```
df['Hour'].value_counts()
```

```
Hour
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
19    1000
Name: count, dtype: int64
```

Analyzing the Data

1) Best selling Product ?

```
df.groupby('Product')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('Product').count()
```

```
                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
Product                                                                                                          
27in 4K Gaming Monitor     500               500         500         500               500    500    500    500   500
27in"                      500               500         500         500               500    500    500    500   500
34in Ultra Wide Monitor    500               500         500         500               500    500    500    500   500
AA Batteries (4-pack)      500               500         500         500               500    500    500    500   500
AAA Batteries (4-pack)     500               500         500         500               500    500    500    500   500
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Flatscr                    500               500         500         500               500    500    500    500   500
Google Phone               500               500         500         500               500    500    500    500   500
iPhone                     500               500         500         500               500    500    500    500   500
LG Dryer                   500               500         500         500               500    500    500    500   500
LG Washing Machine         500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Screen                     500               500         500         500               500    500    500    500   500
Vareebadd phone            500               500         500         500               500    500    500    500   500
Wired                      500               500         500         500               500    500    500    500   500
Wired Headphones           500               500         500         500               500    500    500    500   500
USB-C Charging Cable      1000              1000        1000        1000              1000   1000   1000   1000  1000
```

```
df.groupby('Product').count()['Quantity_Ordered']
```

```
Product
27in 4K Gaming Monitor    500
27in"                     500
34in Ultra Wide Monitor   500
AA Batteries (4-pack)     500
AAA Batteries (4-pack)    500
Apple Airpods Headpho     800
Apple Airpods Headpho     800
Bose SoundSport Head      500
Bose SoundSport Head      500
Flatscr                   500
Google Phone              500
iPhone                    500
LG Dryer                  500
LG Washing Machine        500
Lightning Charging C      500
Lightning Charging C      500
Screen                    500
Vareebadd phone           500
Wired                     500
Wired Headphones          500
USB-C Charging Cable     1000
Name: Quantity_Ordered, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['Product'], df['Quantity_Ordered'])
plt.xlabel('Product')
plt.ylabel('Quantity_Ordered')
plt.xticks(rotation=90)
plt.show()
```

2) Best selling Product ?

```
df.groupby('Product')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('Product').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
Product                                                                                                          
27in 4K Gaming Monitor     500               500         500         500               500    500    500    500   500
27in"                      500               500         500         500               500    500    500    500   500
34in Ultra Wide Monitor    500               500         500         500               500    500    500    500   500
AA Batteries (4-pack)      500               500         500         500               500    500    500    500   500
AAA Batteries (4-pack)     500               500         500         500               500    500    500    500   500
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Flatscr                    500               500         500         500               500    500    500    500   500
Google Phone               500               500         500         500               500    500    500    500   500
iPhone                     500               500         500         500               500    500    500    500   500
LG Dryer                   500               500         500         500               500    500    500    500   500
LG Washing Machine         500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Screen                     500               500         500         500               500    500    500    500   500
Vareebadd phone            500               500         500         500               500    500    500    500   500
Wired                      500               500         500         500               500    500    500    500   500
Wired Headphones           500               500         500         500               500    500    500    500   500
USB-C Charging Cable      1000              1000        1000        1000              1000   1000   1000   1000  1000
>
```

```
df.groupby('Product').count()['Price_Each']
```

```
Product
27in 4K Gaming Monitor    500
27in"                     500
34in Ultra Wide Monitor   500
AA Batteries (4-pack)     500
AAA Batteries (4-pack)    500
Apple Airpods Headpho     800
Apple Airpods Headpho     800
Bose SoundSport Head      500
Bose SoundSport Head      500
Flatscr                   500
Google Phone              500
iPhone                    500
LG Dryer                  500
LG Washing Machine        500
Lightning Charging C      500
Lightning Charging C      500
Screen                    500
Vareebadd phone           500
Wired                     500
Wired Headphones          500
USB-C Charging Cable     1000
Name: Price_Each, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['Product'], df['Price_Each'])
plt.xlabel('Product')
plt.ylabel('Price_Each')
plt.xticks(rotation=90)
plt.show()
```

3) Which product generated maximum revenue

```
df.groupby('Product')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('Product').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
Product                                                                                                          
27in 4K Gaming Monitor     500               500         500         500               500    500    500    500   500
27in"                      500               500         500         500               500    500    500    500   500
34in Ultra Wide Monitor    500               500         500         500               500    500    500    500   500
AA Batteries (4-pack)      500               500         500         500               500    500    500    500   500
AAA Batteries (4-pack)     500               500         500         500               500    500    500    500   500
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Flatscr                    500               500         500         500               500    500    500    500   500
Google Phone               500               500         500         500               500    500    500    500   500
iPhone                     500               500         500         500               500    500    500    500   500
LG Dryer                   500               500         500         500               500    500    500    500   500
LG Washing Machine         500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Screen                     500               500         500         500               500    500    500    500   500
Vareebadd phone            500               500         500         500               500    500    500    500   500
Wired                      500               500         500         500               500    500    500    500   500
Wired Headphones           500               500         500         500               500    500    500    500   500
USB-C Charging Cable      1000              1000        1000        1000              1000   1000   1000   1000  1000
>
```

```
df.groupby('Product').count()['Quantity_Ordered']
```

```
Product
27in 4K Gaming Monitor    500
27in"                     500
34in Ultra Wide Monitor   500
AA Batteries (4-pack)     500
AAA Batteries (4-pack)    500
Apple Airpods Headpho     800
Apple Airpods Headpho     800
Bose SoundSport Head      500
Bose SoundSport Head      500
Flatscr                   500
Google Phone              500
iPhone                    500
LG Dryer                  500
LG Washing Machine        500
Lightning Charging C      500
Lightning Charging C      500
Screen                    500
Vareebadd phone           500
Wired                     500
Wired Headphones          500
USB-C Charging Cable     1000
Name: Quantity_Ordered, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['Product'], df['Quantity_Ordered'])
plt.xlabel('Product')
plt.ylabel('Quantity_Ordered')
plt.xticks(rotation=90)
plt.show()
```

4) Which city had maximum revenue

```
df.groupby('City')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('City').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
City                                                                                                             
Atlanta                    500               500         500         500               500    500    500    500   500
Boston                     500               500         500         500               500    500    500    500   500
Dallas                     500               500         500         500               500    500    500    500   500
Los Angeles                500               500         500         500               500    500    500    500   500
New York City              500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
San Francisco             2000              2000        2000        2000              2000   2000   2000   2000  2000
Seattle                    500               500         500         500               500    500    500    500   500
>
```

```
df.groupby('City').count()['Price_Each']
```

```
City
Atlanta          500
Boston           500
Dallas           500
Los Angeles      500
New York City    500
Portland         500
Portland         500
San Francisco   2000
Seattle          500
Name: Price_Each, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['City'], df['Price_Each'])
plt.xlabel('City')
plt.ylabel('Price_Each')
plt.xticks(rotation=90)
plt.show()
```

5) Which city had maximum revenue

```
df.groupby('City')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('City').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
City                                                                                                             
Atlanta                    500               500         500         500               500    500    500    500   500
Boston                     500               500         500         500               500    500    500    500   500
Dallas                     500               500         500         500               500    500    500    500   500
Los Angeles                500               500         500         500               500    500    500    500   500
New York City              500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
San Francisco             2000              2000        2000        2000              2000   2000   2000   2000  2000
Seattle                    500               500         500         500               500    500    500    500   500
>
```

```
df.groupby('City').count()['Sales']
```

```
City
Atlanta          500
Boston           500
Dallas           500
Los Angeles      500
New York City    500
Portland         500
Portland         500
San Francisco   2000
Seattle          500
Name: Sales, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['City'], df['Sales'])
plt.xlabel('City')
plt.ylabel('Sales')
plt.xticks(rotation=90)
plt.show()
```

6) Which city had maximum revenue

```
df.groupby('City')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('City').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
City                                                                                                             
Atlanta                    500               500         500         500               500    500    500    500   500
Boston                     500               500         500         500               500    500    500    500   500
Dallas                     500               500         500         500               500    500    500    500   500
Los Angeles                500               500         500         500               500    500    500    500   500
New York City              500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
San Francisco             2000              2000        2000        2000              2000   2000   2000   2000  2000
Seattle                    500               500         500         500               500    500    500    500   500
>
```

```
df.groupby('City').count()['Sales']
```

```
City
Atlanta          500
Boston           500
Dallas           500
Los Angeles      500
New York City    500
Portland         500
Portland         500
San Francisco   2000
Seattle          500
Name: Sales, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['City'], df['Sales'])
plt.xlabel('City')
plt.ylabel('Sales')
plt.xticks(rotation=90)
plt.show()
```

7) Which city had maximum revenue

```
df.groupby('City')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('City').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
City                                                                                                             
Atlanta                    500               500         500         500               500    500    500    500   500
Boston                     500               500         500         500               500    500    500    500   500
Dallas                     500               500         500         500               500    500    500    500   500
Los Angeles                500               500         500         500               500    500    500    500   500
New York City              500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
Portland                   500               500         500         500               500    500    500    500   500
San Francisco             2000              2000        2000        2000              2000   2000   2000   2000  2000
Seattle                    500               500         500         500               500    500    500    500   500
>
```

```
df.groupby('City').count()['Sales']
```

```
City
Atlanta          500
Boston           500
Dallas           500
Los Angeles      500
New York City    500
Portland         500
Portland         500
San Francisco   2000
Seattle          500
Name: Sales, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['City'], df['Sales'])
plt.xlabel('City')
plt.ylabel('Sales')
plt.xticks(rotation=90)
plt.show()
```

8) Which product generated maximum revenue

```
df.groupby('Product')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('Product').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
Product                                                                                                          
27in 4K Gaming Monitor     500               500         500         500               500    500    500    500   500
27in"                      500               500         500         500               500    500    500    500   500
34in Ultra Wide Monitor    500               500         500         500               500    500    500    500   500
AA Batteries (4-pack)      500               500         500         500               500    500    500    500   500
AAA Batteries (4-pack)     500               500         500         500               500    500    500    500   500
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Apple Airpods Headpho      800               800         800         800               800    800    800    800   800
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Bose SoundSport Head       500               500         500         500               500    500    500    500   500
Flatscr                    500               500         500         500               500    500    500    500   500
Google Phone               500               500         500         500               500    500    500    500   500
iPhone                     500               500         500         500               500    500    500    500   500
LG Dryer                   500               500         500         500               500    500    500    500   500
LG Washing Machine         500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Lightning Charging C       500               500         500         500               500    500    500    500   500
Screen                     500               500         500         500               500    500    500    500   500
Vareebadd phone            500               500         500         500               500    500    500    500   500
Wired                      500               500         500         500               500    500    500    500   500
Wired Headphones           500               500         500         500               500    500    500    500   500
USB-C Charging Cable      1000              1000        1000        1000              1000   1000   1000   1000  1000
>
```

```
df.groupby('Product').count()['Sales']
```

```
Product
27in 4K Gaming Monitor    500
27in"                     500
34in Ultra Wide Monitor   500
AA Batteries (4-pack)     500
AAA Batteries (4-pack)    500
Apple Airpods Headpho     800
Apple Airpods Headpho     800
Bose SoundSport Head      500
Bose SoundSport Head      500
Flatscr                   500
Google Phone              500
iPhone                    500
LG Dryer                  500
LG Washing Machine        500
Lightning Charging C      500
Lightning Charging C      500
Screen                    500
Vareebadd phone           500
Wired                     500
Wired Headphones          500
USB-C Charging Cable     1000
Name: Sales, dtype: int64
```

```
plt.figure(figure_size=(12,6))
plt.bar(df['Product'], df['Sales'])
plt.xlabel('Product')
plt.ylabel('Sales')
plt.xticks(rotation=90)
plt.show()
```

9) Sales Analysis

```
df.groupby('Order_Date')
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7f8d0a0a0a0a>
```

```
df.groupby('Order_Date').count
```

```
<bound method DataFrame.count of                  Order_ID  Quantity_Ordered  Price_Each  Order_Date  Purchase_Address  Month  Sales   City  Hour
Order_Date                                                                                                       
01/01/2019 00:00           500               500         500         500               500    500    500    500   500
01/01/2019 00:00           500               500         500         500               500    500    500    500   500
01/01/2019 00:00           500               500         500         500               500    500    500    500   500
```